MILITARY SPECIFICATION

SUPPRESSOR, ELECTRICAL TRANSIENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of a shunt transient suppressor assembly designated as Suppressor, Electrical Transient DL-SM-B-627930. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-Handling, Wood, (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-20	Bag, Cotton, Mailing
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade)
PPP-P-291	Paperboard, Wrapping, Cusioning
PPP-P-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape; Pressure Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116	Preservation-Packaging, Methods of
MIL-S-901	Shock Tests H.I. (High Impact) Shipboard Machinery, Equipments and Systems, Requirements for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-B-43014	Box, Water Resistant Paperboard Folding, Set-Up and Metal Stayed

STANDARDS

FEDERAL

FED-STD-595	Colors

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

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DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-627930 Suppressor, Installation Kit

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. - The equipment shall be constructed in accordance with Drawing DL-SM-B-627930.

3.3 Bounce preconditioning. - The equipment shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning specified in 4.6.

3.4 Electrical requirements.

3.4.1 Voltage limitation. - When tested in accordance with 4.7.1 the operating voltage shall be 36+0, -3 Volts at a suppression element temperature of 75°F.

3.4.2 Pulsing.- When tested in accordance with 4.7.2, the voltage rise at the input terminals of the suppressor during a 100 +0, -5 Volt, 50 millisecond pulse shall not exceed 8 Volts and the current droop shall not exceed 20 amperes. The suppressor fuse shall not blow during this test. The suppressor voltage during each pulse shall not fall below the voltage measured in 3.4.1.

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3.4.2.1 Pulse circuit. - The pulsing circuit shall be capable of delivering five successive 50 millisecond pulses at ten second intervals of 125 amperes minimum when loaded with a 0.30 +0.00, -0.05 ohms resistive load. The pulse rise time shall be not longer than one millisecond. The pulse droop shall be less than 10 amperes at 50 milliseconds. The total internal and external impedance of the voltage source and pulser shall be 0.50 +0.00 ohms (See 4.7.2.1)
-0.05

3.4.3 Battle override. - When tested in accordance with 4.7.3, the suppressor battle override indicator lamp shall operate without failure.

3.4.4 Power dissipation. - When tested in accordance with 4.7.4, the suppressor power dissipation shall not cause suppressor failure and there shall be no evidence of mechanical damage and meet the requirements of 3.4.1 following the test.

3.4.5 Blocking voltage. - When tested in accordance with 4.7.5, the suppressor shall show no evidence of electrical or mechanical damage. The measured leakage current shall not exceed 2 milliamperes when a reverse potential of 50 Volts dc is applied in room ambient of 75 $\pm 5^{\circ}$F.

3.5 Service conditions. - The equipment shall meet the following service conditions.

3.5.1 Temperature.

3.5.1.1 High temperature. - The unit shall not be damaged when subjected to a storage temperature of 160°F. The unit shall meet the electrical requirements of 3.4.1 and 3.4.2 at temperature and times specified when tested in accordance with 4.8.1.1 and 3.4.3 and 3.4.5 after testing.

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3.5.1.2 Low temperature. - The unit shall not be damaged when subjected to a storage temperature of -65°F. The unit shall meet the electrical requirements of 3.4.1 and 3.4.2 at temperatures and times specified when tested in accordance with 4.8.1.2 and 3.4.3 and 3.4.5 after testing.

3.5.2 Humidity. - The suppressor shall exhibit no physical damage such as corrosion, rust, blistering, swelling, deterioration, and shall meet the electrical requirements of 3.4.1, 3.4.2, and 3.4.3 and 3.4.5 when tested in accordance with 4.8.2.

3.5.3 Altitude. - The equipment shall not be damaged and shall meet the electrical requirements of 3.4.1 and 3.4.2 at times specified when tested in accordance with 4.8.3, and 3.4.3 and 3.4.5 after testing.

3.5.4 Vibration. - When tested in accordance with 4.8.4 and with the suppressor conducting 10 milliamperes (maximum) of current no intermittent operation of the suppressor shall occur. The unit shall show no mechanical damage and shall meet the requirements of 3.4.1, 3.4.2, 3.4.3, and 3.4.5 after testing.

3.5.5 Shock, ballistic. - The equipment shall show no failure of its attachment and shall remain in place when tested in accordance with 4.8.5. However, bending and distortion shall be permitted. The equipment shall meet the electrical requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.5 after testing.

3.5.6 Sand and dust. - The equipment shall show no mechanical damage and shall meet the requirements of 3.4.1 and 3.4.2, 3.4.3 and 3.4.5 after testing in accordance with 4.8.6.

3.5.7 Immersion. - The equipment shall be capable of being immersed as described in 4.8.7 without any evidence of water leakage.

3.5.8 Salt fog. - After the salt fog test of 4.8.8, the suppressor when examined visually with the aid of a 10-power magnifier shall show no evidence of degredation, such as flaking, pitting, blistering or loosening of finish or metal surface; or exfoliation (see 6.8) of metal. Electrical operation is not required.

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3.5.9 Fungus. - After the fungus test of 4.8.9, the suppressor shall be visually examined using a 10-power magnifier. The suppressor shall show no more than six minute unrelated spots each no greater than 0.015 square inch in area, of sparse microbial growth (6.8) as evidenced by growth colonization (6.8) (which includes branching (6.8) and sporulation (6.8) on or within each cubic foot or fraction thereof of equipment assembly volume. Isolated instances of partial tubular germination (6.8) shall not be included in this evaluation.

3.6 Finish, protective. - Equipment, except the heat sink, shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be green color (olive drab), semi-gloss enamel, except the heat sink which shall be finished per MIL-A-8625, Type II, Class 2, olive drab.

3.6.1 Color. - The final paint film on Type I surfaces shall be smooth semi-gloss, green, color conforming to Nr. X24087 of Federal Standard FED-STD-595. (See 4.4).

3.7 Marking. - Marking shall conform to Specification MIL-M-13231. Panel marking shall be Group I as described in that specification. (See 4.4).

3.7.1 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment. (See 4.4).

3.7.2 Serial numbers. - Serial numbers are not required.

3.8 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.9)

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3.9 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, Protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Soldering.

(5) Cleaning.

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Mechanical operation of controls.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unlessotherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. - Inspection requirements specified herein are classified as follows:

(a) First Article Inspection (Does not include preparation for delivery.) (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 First article. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. - The contractor shall furnish four (4) first article units of the complete suppressor and one pulse circuit.

4.3.2 First article inspection. - The first article inspection shall consist of the inspection specified in Table I, and shall be performed in the order specified.

4.3.3 First article data. - The first article test plan and test reports shall be as required in the contract or purchase order.

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TABLE I - FIRST ARTICLE INSPECTION

INSPECTIONS	REQT. PARA.	TEST PARA.	SUPPRESSOR UNIT 1	UNIT 2	UNIT 3	UNIT 4	PULSE CIRCUIT
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units				
2. Bounce pre-conditioning	3.3	4.6	Inspections to be performed on all units				
3. Group A Inspection	See Table II		Inspection to be performed on all units				
4. Group B Inspection	See Table III		Inspection to be performed on all units				
5. Environmental							
High Temperature	3.5.1.1.	4.8.1.1	1				
Low Temperature	3.5.1.2	4.8.1.2	2				
Humidity	3.5.2	4.8.2	3				
Shock, Ballistic	3.5.5	4.8.5	4				
Altitude	3.5.3	4.8.3		1			
Salt Fog	3.5.8	4.8.8		2			
Vibration	3.5.4	4.8.4			1		
Sand-Dust	3.5.6	4.8.6			2		
Immersion	3.5.7	4.8.7			3		
Fungus	3.5.9	4.8.9				1	
6. Pulse Circuit	3.4.2.1	4.7.2.1					1

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4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Finish	3.6
Color	3.6.1
Marking	3.7

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.6, 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in any order satisfactory to the Government.

TABLE II-GROUP A INSPECTION

INSPECTION	REQT. PARA.	INSP. PARA.	AQL MAJOR	AQL MINOR
Visual & Mechanical	3.9	4.10	1%	4.0%
Voltage Limitation	3.4.1	4.7.1)	1%	*
Pulsing	3.4.2	4.7.2)	For	*
Battle Override	3.4.3	4.7.3)	the	*
Blocking Voltage	3.4.5	4.7.5)	Group	*

* All electrical defects are major.

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4.5.2 Group B Inspection. - Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. - The Group B sampling plans, for the AQL's listed in Table III, shall be as follows:

AQL's	Inspection Level
1.0%	S-4
4.0%	S-4

4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in an order satisfactory to the Government.

TABLE III - GROUP B INSPECTION

INSPECTION	REQT. PARA.	INSP. PARA.	AQL
Power Dissipation	3.4.4	4.7.4	1.0%
Interchangeability	3.8	4.9	4.0%

4.5.3 Group C inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

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TABLE IV - GROUP C INSPECTION

INSPECTION	REQ. PARA.	INSP. PARA.
Group I		
Immersion	3.5.7	4.8.7
Group IIA		
High Temperature	3.5.1.1	4.8.1.1
Low Temperature	3.5.1.2	4.8.1.2
Vibration	3.5.4	4.8.4
Group IIB		
Altitude	3.5.3	4.8.3
Salt Fog	3.5.8	4.8.8
Shock, Ballistic	3.5.5	4.8.5
Group IIIA		
Humidity	3.5.2	4.8.2
Sand and Dust	3.5.6	4.8.6
Group IIIB		
Fungus	3.5.9	4.8.9
Group IV		
Pulse Circuit Test	3.4.2.1	4.7.2.1

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4.5.3.1 Sampling for Group C inspections

4.5.3.1.1 Group I, Table IV. - Two samples shall be selected at random, from each 100 equipments produced and subjected to each inspection specified. The first two samples shall be selected at the start of the contract from the first quality conformance lot. (See 4.5)

4.5.3.1.2 Groups IIA and IIB. - Two samples shall be selected at random for each sub-group, from each 500 equipments produced and subjected to each inspection specified. The first two samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5)

4.5.3.1.3 Groups IIIA and IIIB. - One sample shall be selected at random for each sub-group, from each 1000 equipments produced and subjected to the inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5)

4.5.3.1.4 Group IV. - The pulse circuit test shall be performed once for each 1000 equipments produced.

4.5.3.2 Order of inspection within Group C. - Group C inspection shall be performed in the order shown in each subgroup in Table IV.

4.5.3.3 Group C failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C inspection, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Bounce preconditioning. - Each unit of each lot shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. (See 3.3).

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4.7 Electrical tests.

4.7.1 Voltage limitation. - Using standard laboratory test instruments, the voltage measured at the terminals of a mating connector to the suppressor shall meet the requirements of 3.4.1 when conducting a current of 1 ampere dc ±10 percent. Voltage measurement shall be made at approximately 2 seconds after application of load. For suppression element temperatures of less than 75°F subtract 17.2 millivolts/°F from the voltage as measured at 75°F, and for suppression element temperatures of more than 75°F add 17.2 millivolts/°F to the voltage as measured at 75°F.

4.7.2 Pulsing. - The suppressor shall be pulsed 5 successive times at 10 second intervals with a 100 +0, -5 Volt, 50 millisecond pulse (as measured without the suppressor connected) from a power supply having a 0.50 +0, -0.05 ohm total source resistance and capable of supplying 200 amperes for 50 milliseconds. Storage batteries, with series resistance as required to obtain the total source resistance of 0.50 +0, -0.05 ohm, without the suppressor connected, can be used. The voltage at the suppressor terminals during each pulse shall be measured from 1 millisecond, ±10 percent, after the start of the pulse to the end of the pulse and shall be in accordance with the requirements of 3.4.2. The current droop during the pulse shall meet the requirements of 3.4.2. The suppressor shall meet the requirements of 3.4.1 following the test.

4.7.2.1 Pulse circuit. - The pulse circuit shall be tested to assure conformance to the requirements of 3.4.2.1. The total internal and external resistance of the voltage source and pulser shall be determined as follows:

1. With the suppressor removed from the pulse test circuit, the power supply shall be adjusted to obtain a resistance load current of approximately 120 amperes (50 millisecond pulse).

2. The load voltage and current shall be measured.

3. With the resistance load removed, and the power supply setting undisturbed, measure the open circuit voltage at the load terminals.

4. Determine the difference in load voltage measurements made in (2) and (3) and divide the difference by the measured load current to obtain the total source resistance.

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4.7.3 Battle override. - The battle override indicator lamp shall be removed from the suppressor for this test. Using standard laboratory test instruments, a voltage of not less than 45 volts dc shall be applied to the lamp for not less than 1 second. The power supply used shall have an output current rating of at least 0.75 amperes dc. The requirements of 3.4.3 shall be met.

4.7.4 Power dissipation test. - Using standard laboratory test instrument, 70 watts dc shall be dissipated continuously for a total test period of not less than 45 minutes. The test shall be performed with the suppressor installed in the MT-1898/VRC mounting in room ambient of 75 ±5°F. The requirements of 3.4.4 shall be met following the test.

4.7.5 Blocking voltage. - Using standard laboratory test instruments, reverse polarity voltage of 50 ±5 volts shall be applied to the suppressor and leakage current measured. The test shall be performed in room ambient of 75±5°F. The requirements of 3.4.5 shall be met.

4.8 Service conditions. - The suppressor shall be tested in accordance with the following service conditions:

4.8.1 Temperature.

4.8.1.1 High Temperature. - The equipment shall be subjected to Test Method 501, Procedure I, of MIL-STD-810. The storage temperature shall be +160°F for 48 hours and the operating temperature shall be +155°F. The unit shall meet the requirements of 3.5.1.1.

4.8.1.2 Low temperature. - The equipment shall be subjected to Test Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -65°F for 48 hours and the operating temperature shall be -40°F. The unit shall meet the requirements of 3.5.1.2.

4.8.2 Humidity. - The equipment shall be subjected to the test specified in Method 507, Procedure III, of MIL-STD-810 and shall meet the requirements of 3.5.2. Electrical measurements shall be made only during step 11 of Procedure III.

4.8.3 Altitude. - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810 and shall meet the requirements of 3.5.3.

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4.8.4 Vibration.

4.8.4.1 Mounting. - The suppressor shall be installed in the MT-1029/VRC mounting. Equivalent radio mass shall be secured to the MT-1029/VRC mounting prior to the test. The equivalent radio mass shall have a total weight of 54 pounds and a center of gravity located 6 inches from the front of the mounting, 7 inches from the left side and 3 inches above the top plate of the mounting. During vibration, a zener current of 10 milliamperes (maximum) shall be maintained. In Part 1, the isolation mounts shall be in place. In Part 2, the isolation mounts shall be blocked. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible the test load shall be distributed uniformly to the vibrator exciter table in order to reduce to the smallest amount, effects of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test item input control or response. The input control sensing device(s) shall be rigidly attached to the vibration table or the immediate structure, "if used", at or near as possible to the attachment point(s) of the test item.

4.8.4.2 Cycling. - The test item shall be vibrated along each of its three (3) mutually perpendicular axis in accordance with the Test Levels, Frequency Range, and Time Schedules outlined below. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range. The test shall be comprised of two (2) parts.

4.8.4.2.1 Part I. (Isolators in place)

Test Level Vibration Frequency of 5.5 Hz to 30 Hz, at an acceleration level of 1.5 G's, 30 Hz to 49 Hz at a double amplitude of .033 inches, 48 Hz to 500 Hz, at an acceleration level of 4.0 G's.

Frequency Range

5.5 - 500 - 5.5 Hz

Time schedule: 3 Hours Per Axis.

Sweep rate: 5 - 500 - 5 Hz in Fifteen (15) minutes.

At the conclusion of the above test, visually inspect the test item for any evidence of mechanical damage.

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4.8.4.2.2 Part II. (Hard mounted)

Test Level Vibration frequency of 5.5 Hz to 38 Hz at an acceleration level of 1.5 g's, 38 Hz to 50 Hz, at a double amplitude of 0.02 inch, 50 Hz to 200 Hz at an acceleration level of 2.5 g's..

Frequency Range 5.5 Hz to 200 to 5.5 Hz

Time schedule: 84 minutes per axis

Sweep rate: 5 - 200 - 5 Hz in 12 minutes.

The suppressor shall meet the requirements of 3.5.4.

4.8.5 Shock, ballistic. - The suppressor shall be mounted in the MT-1029/VRC mounting. The test shall be performed using the "Shock Testing Machine For Lightweight Equipment" shown in MIL-S-901. Equivalent radio mass as defined in 4.8.4 shall be mounted in the MT-1029/VRC mounting prior to the test. The test shall consist of a total of 9 blows; one each at 1, 3, and 5 feet hammer drops on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the side blows, equivalent rotation of the equipment under test is permissible. The suppressor shall meet the requirements of 3.5.5.

4.8.6 Sand and dust. - The equipment shall be subjected to the test of Method 510, Procedure I of MIL-STD-810. The equipment shall meet the requirements of 3.5.6 following the test.

4.8.7 Immersion. - The equipment shall be subjected to the test of Method 512, Procedure I, of MIL-STD-810. The equipment shall meet the requirements of 3.5.7 following the test.

4.8.8 Salt fog. - The equipment shall be subjected to the salt fog test specified in Method 509, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the test facility. The equipment shall meet the requirements of 3.5.8 following the test.

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4.8.9 Fungus. - The suppressor shall be subjected to the fungus test specified in Method 508, Procedrue I, of MIL-STD-810. There shall be abundant growth colonization (6.8) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the suppressor is permitted for 72 hours prior to the fungus test. Handling before, during, and after the test procedure shall be accomplished without contamination of equipment. Inability of the suppressor to meet the requirements of 3.5.9 shall constitute failure of this test.

4.9 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.8. When listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.10 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in MIL-STD-252.

4.11 Rough handling test (preparation for delivery). - When the rough handling test in accordance with MIL-P-116 is required by contract (see 6.2 c) the unit shall be subjected to the tests of 4.7.1 thru 4.7.5 to determine freedom from malfunction caused by the rough handling.

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2 b).

5.1.1 Level A.

5.1.1.1 Cleaning. - Suppressor, electrical transient shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. - Suppressor, electrical transient shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Screws and lockwashers. - Screws and lockwashers for each suppressor shall be packaged together Method III as follows: Place the screws and lockwashers within the smallest size cotton bag conforming to PPP-B-20 and secure the bag with the drawstrings.

5.1.1.4.2 Silicon grease. - Tubes of silicon grease will require no packaging.

5.1.1.4.3 Heat sink and guard plate. Each heat sink and guard plate shall be individually packaged Method III as follows: Cushion each heat sink and guard plate by wrapping in paperboard conforming to PPP-P-291, Type I. Secure the wrap with tape conforming to PP-T-45, Type III, Grade A.

5.1.1.4.4 Suppressor, electrical transient. - Each suppressor shall be individually packaged Method III as follows: Package each item individually as specified in 5.1.1.4.3. Place the packaged suppressor and the screws, lockwashers, silicon grease and heat sink, packaged as specified in 5.1.1.4.1 through 5.1.1.4.3, within a close-fitting paperboard box conforming to MIL-B-43014, Form 1, Style II, Type A. Fill all voids with paperboard conforming to PPP-P-291, Type I, to prevent movement during handling and shipment. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level C. - Suppressor, Electrical transient shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2b).

5.2.1 Level A.

5.2.1.1 Consolidation. - A quantity of suppressor, electrical, transient packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF class weather resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A or Shrink Film.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48x24 inches or more and the weight exceeds 100 pounds, 3x4 inch inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and

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5.2.1.3 (Cont'd) facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class A.

5.2.2 Level B.

5.2.2.1 Consolidation. - A quantity of SUPPRESSOR, ELECTRICAL, TRANSIENT, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. - A quantity of SUPPRESSOR, ELECTRICAL TRANSIENT, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - The transient suppressor shall operate with tactical radios operating from 100 to 400 ampere, 28 volt dc, vehicular power supplies. The suppressor is intended to be used with tactical radios (AN/VRC-12, AN/VRC-46, AN/VRC-53, AN/VRC-64 and their configurations when operated from vehicular power supplies) in all military vehicles. It shall be packaged for mounting in the MT-1029/VRC and MT-1898/VRC vehicular radio mountings.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level of preservation and packaging and packing. (See Section 5).

c. When rough handling and functional tests are required.

d. First article inspection:

Four sample units of each item cited in the sectionl are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Quantity of Government furnished test equipment, Mounts MT-1029/VRC, MT-1898/VRC, and replacement shock isolators FSN: 5340-874-3590 for MT-1029 mount.

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6.3 Color. - The color chip furnished by the procuring agency (See 3.7.1) will be color chip no. X-24087, and may be obtained upon request to Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Grounds, Maryland.

6.4 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3).

6.5 Location of inspection. - It is desirable that the inspection be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, constructive methods, etc.

6.7.1 Service conditions. - Continuous use for a period of 5 years, under the world-wide environmental conditions specified in 3.5.1 through 3.5.9 with a duty cycle of 8 hours on and 4 hours off and with no more than normal maintenance and replacement of parts.

6.7.2 Safety. - The equipment was designed to meet the pertinent safety requirements of requirement 1 of MIL-STD-454.

6.7.3 Human factors. - The equipment was designed to meet the pertinent requirements of MIL-STD-1472.

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6.8 Definitions.

6.8.1 Branching. - Branching is connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.8.2 Exfoliation. - Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.8.3 Growth colonization. - Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.4 Microbial growth. - Microbial growth is the growth of very minute organisms when present in large numbers may provide a colony visible to the naked eye.

6.8.5 Sporulation. - Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.8.6 Tubular germination. - Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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Project Nr. 5915-A056

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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